Linear/Matrix Algebra By Hand Questions

12 Points Possible

Statistics 469: Analysis of Correlated Data

Try the following practice problems **by hand and show your work.**

1. Compute the following:
    a. $\mathbf{A} + \mathbf{B}$
    b. $\mathbf{A} - \mathbf{B}$
    c. $\mathbf{AB}$
    d. $\mathbf{BA}$

    where

$$\boldsymbol{A} = \begin{pmatrix} 1 & 3 \\ 5 & 2 \end{pmatrix}, \quad \boldsymbol{B} = \begin{pmatrix} 4 & 9 \\ 2 & 6 \end{pmatrix}$$

2. If possible, compute the following. If non-conformable just put "non-conformable".
    a. $\mathbf{A} + \mathbf{B}$
    b. $\mathbf{A} - \mathbf{B}$
    c. $\mathbf{AB}$
    d. $\mathbf{BA}$

    where

$$\mathbf{A} = \begin{pmatrix} 9 & 1 \\ 1 & 4 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 6 & 5 \\ 3 & 1 & 8 \end{pmatrix}$$

3. Using the properties of inverses and transposes that I listed above, show that the following is true for any invertible, square matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ of the same size:

$$[(\mathbf{A} + \mathbf{B})'\mathbf{D}^{-1}\mathbf{C}^{-1}]^{-1} = \mathbf{CD}(\mathbf{A}' + \mathbf{B}')^{-1}$$

4. Define

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix},$$
$$\mathbf{B} = \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix}$$

    where each $\mathbf{A}_{ij}$ and $\mathbf{B}_{lm}$ is a $p \times p$ invertible matrix. Write out the following in terms of the sub-matrices:

    a. $\mathbf{AB}$
    b. $\mathbf{A}'$

5. If

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}$$

Find $dq(\boldsymbol{x})/d\boldsymbol{x}$ where $q(\boldsymbol{x}) = \boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}$.